Installing, upgrading, and removing Unified Manager software

Active IQ Unified Manager 9.11

NetApp
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Table of Contents

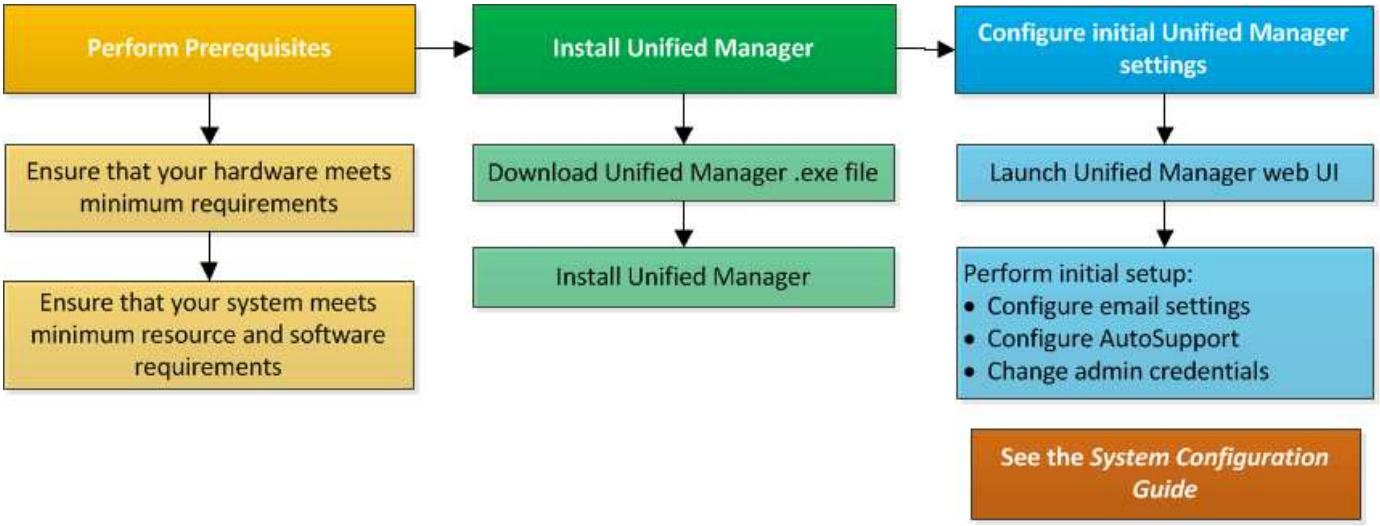
Installing, upgrading, and removing Unified Manager software . . . 1
- Overview of the installation process . . . 1
- Installing Unified Manager on Windows . . . 1
- Changing the JBoss password . . . 5
- Supported upgrade path for Unified Manager versions . . . 5
- Upgrading Unified Manager . . . 6
- Upgrading third-party products . . . 7
- Restarting Unified Manager . . . 8
- Uninstalling Unified Manager . . . 9

Installing, upgrading, and removing Unified Manager software

You can install Unified Manager software, upgrade to a newer version of software, or remove the Unified Manager application.

Overview of the installation process

The installation workflow describes the tasks that you must perform before you can use Unified Manager.



Installing Unified Manager on Windows

It is important that you understand the sequence of steps to download and install Unified Manager on Windows.

Installing Unified Manager

You can install Unified Manager to monitor and troubleshoot data storage capacity, availability, performance, and protection issues.

What you'll need

- The system on which you plan to install Unified Manager should meet the system and software requirements.

  Hardware system requirements

  Windows software and installation requirements

  > Starting with Unified Manager 9.5, OpenJDK is provided in the installation package and installed automatically. Oracle Java is not supported starting with Unified Manager 9.5.

- You should have Windows administrator privileges. Ensure that your user name does not begin with an exclamation mark “!”. Installation of Unified Manager might fail if the user name of user running the installation begins with “!”.
- You should have a supported web browser.
- The Unified Manager maintenance user password should be between 8 and 20 characters, should contain upper-case letters or lower-case letters, numerals, and special characters.
- The following special characters are not allowed in the password string for the maintenance user or for the MySQL root user: " ' ` % , = & < > | ^ \ / ( ) [ ] ;:

  The following special characters are allowed: ~ ! @ # $ * - ? . + { }

Steps

1. Log in to Windows using the default local administrator account.
2. Log in to the NetApp Support Site, and navigate to the Download page for Unified Manager:

   NetApp Support Site

3. Select the required version of Unified Manager and accept the end-users license agreement (EULA).
4. Download the Unified Manager Windows installation file to a target directory on the Windows system.
5. Navigate to the directory where the installation file is located.
6. Right-click and run the Unified Manager installer executable (.exe) file as an administrator.

   Unified Manager detects missing or pre-installed third-party packages and lists them. If the required third-party packages are not installed in the system, Unified Manager installs them as part of the installation.

7. Click **Next**.
8. Enter the user name and password to create the maintenance user.
9. In the Database Connection wizard, enter the MySQL root password.
10. Click **Change** to specify a new location for the Unified Manager installation directory and MySQL data directory.

    If you do not change the installation directory, Unified Manager is installed in the default installation directory.

11. Click **Next**.
12. In the Ready to Install Shield wizard, click **Install**.
13. After the installation is complete, click **Finish**.
14. If you have an active anti-virus software installed on your Windows system, manually exclude the following paths from anti-virus scan after your installation is complete:
    - Unified Manager data directory
    - Unified Manager installation directory
    - MySQL data directory

The installation creates multiple directories:

- Installation directory

This is the root directory for Unified Manager, which you specified during installation. Example: `C:\Program Files\NetApp\`

- MySQL data directory

  This is the directory where the MySQL databases are stored, which you specified during installation. Example: `C:\ProgramData\MySQL\MySQLServerData\`

- Java directory

  This is the directory where OpenJDK is installed. Example: `C:\Program Files\NetApp\JDK\`

- Unified Manager application data directory (appDataDir)

  This is the directory where all the application-generated data is stored. This includes logs, support bundles, backup, and all other additional data. Example: `C:\ProgramData\NetApp\OnCommandAppData\`

You can access the web UI to perform the initial setup of Unified Manager, as described in the Configuring Active IQ Unified Manager.

Performing an unattended installation of Unified Manager

You can install Unified Manager without user intervention by using the command-line interface. You can complete the unattended installation by passing the parameters in key-value pairs.

Steps

1. Log in to the Windows command-line interface by using the default local administrator account.
2. Navigate to the location where you want to install Unified Manager, and choose one of the following options:

Option	Instructions

If third-party packages are pre-installed	`ActiveIQUnifiedManager-x.y.exe /V"MYSQL_PASSWORD=mysql_password INSTALLDIR=\"Installation directory\" MYSQL_DATA_DIR=\"MySQL data directory\" MAINTENANCE_PASSWORD=maintenance_password MAINTENANCE_USERNAME=maintenance_username /qn /l*v CompletePathForLogFile"`
	
	Example:
	
	`ActiveIQUnifiedManager.exe /s /v"MYSQL_PASSWORD=netapp21! INSTALLDIR=\"C:\Program Files\NetApp\" MYSQL_DATA_DIR=\"C:\ProgramData\MYSQL\MySQLServer\" MAINTENANCE_PASSWORD=* MAINTENANCE_USERNAME=admin /qn /l*v C:\install.log"`

If third-party packages are not installed	`ActiveIQUnifiedManager-x.y.exe /V"MYSQL_PASSWORD=mysql_password INSTALLDIR=\"Installation directory\" MYSQL_DATA_DIR=\"MySQL data directory\" MAINTENANCE_PASSWORD=maintenance_password MAINTENANCE_USERNAME=maintenance_username /qr /l*v CompletePathForLogFile"`
	
	Example:
	
	`ActiveIQUnifiedManager.exe /s /v"MYSQL_PASSWORD=netapp21! INSTALLDIR=\"C:\Program Files\NetApp\" MYSQL_DATA_DIR=\"C:\ProgramData\MYSQL\MySQLServer\" MAINTENANCE_PASSWORD=* MAINTENANCE_USERNAME=admin /qr /l*v C:\install.log"`

The `/qr` option enables quiet mode with a reduced user interface. A basic user interface is displayed, which shows the installation progress. You are not prompted for inputs. If third-party packages such as JRE, MySQL, and 7zip are not pre-installed, you should use the `/qr` option. Installation fails if the `/qn` option is used on a server where third-party packages are not installed.

The `/qn` option enables quiet mode with no user interface. No user interface or details are displayed during installation. You should not use the `/qn` option when third-party packages are not installed.

3. Log in to the Unified Manager web user interface by using the following URL:

```
https://IP address
```

Changing the JBoss password

You can reset the instance-specific JBoss password that is set during installation. You can reset the password optionally, in case your site requires this security capability to override the Unified Manager installation setting. This operation also changes the password JBoss uses to access MySQL.

What you'll need

- You should have Windows admin privileges for the system on which Unified Manager is installed.
- You should have the password for the MySQL root user.
- You should be able to access the NetApp-provided `password.bat` script in the directory

  `C:\Program Files\NetApp\essentials\bin`.

Steps

1. Log in as the admin user on the Unified Manager host machine.
2. Use the Windows Services console to stop the following Unified Manager services:
   - NetApp Active IQ Acquisition Service (Ocie-au)
   - NetApp Active IQ Management Server Service (Oncommandsvc)
3. Launch the `password.bat` script to begin the password change process:

   ```
   C:\Program Files\NetApp\essentials\bin> password.bat resetJBossPassword
   ```

4. When prompted, enter the MySQL root user password.
5. When prompted, enter the new JBoss user password, and then enter it again for confirmation.

   Note that the password should be between 8 and 16 characters, and must contain at least a digit, an upper case and lower case characters, and at least one of these special characters:

   ```
   !@%^*-_+=[]:<>.?/~
   ```

6. When the script completes, start the Unified Manager services by using the Windows Services console:
   - NetApp Active IQ Management Server Service (Oncommandsvc)
   - NetApp Active IQ Acquisition Service (Ocie-au)
7. After all of the services are started, you can log in to the Unified Manager UI.

Supported upgrade path for Unified Manager versions

Active IQ Unified Manager supports a specific upgrade path for each version.

Not all versions of Unified Manager can perform an in-place upgrade to later versions. The Unified Manager upgrades are limited to an N-2 model, meaning an upgrade can only be performed within the next 2 releases on all platforms. For example, you can perform an upgrade to Unified Manager 9.11 from Unified Manager 9.9

and 9.10 only.

If you are running a version that is earlier than the supported versions, your Unified Manager instance will need to be upgraded to one of the supported versions first, and then upgraded to the current version.

For example, if your installed version is OnCommand Unified Manager 9.5 and you want to upgrade to the latest release Active IQ Unified Manager 9.11, you follow a sequence of upgrades.

Sample upgrade path:

1. Upgrade OnCommand Unified Manager 9.5 → Active IQ Unified Manager 9.7.
2. Upgrade 9.7 → 9.9.
3. Upgrade 9.9 → 9.11.

For more information on the upgrade path matrix, see this knowledge base (KB) article.

Upgrading Unified Manager

You can upgrade Unified Manager 9.9 or 9.10 to 9.11 by downloading and running the installation file on the Windows platform.

What you'll need

- The system on which you are upgrading Unified Manager should meet the system and software requirements.

  Hardware system requirements

  Windows software and installation requirements

  > Starting with Unified Manager 9.5, OpenJDK is provided in the installation package and installed automatically. Oracle Java is not supported starting with Unified Manager 9.5.

  > Ensure that Microsoft .NET 4.5.2 or greater is installed on your system before starting the upgrade.

- MySQL Community Edition is automatically upgraded during Unified Manager upgrade. If the installed version of MySQL on your system is earlier than 8.0.28, the Unified Manager upgrade process automatically upgrades MySQL to 8.0.28. You must not run a standalone upgrade of an earlier version of MySQL to 8.0.27.
- You should have Windows administrator privileges. Ensure that your user name does not begin with an exclamation mark "!". Installation of Unified Manager might fail if the user name of user running the installation begins with "!".
- You should have valid credentials to log in to the NetApp Support Site.
- To avoid data loss, you should have created a backup of the Unified Manager machine in case there is an issue during the upgrade.
- You should have adequate disk space available to perform the upgrade.

  The available space on the installation drive should be 2.5 GB larger than the size of the data directory. The upgrade stops and displays an error message indicating the amount of space to be added if there is not enough free space.

- During upgrade, you may be prompted to confirm whether you want to keep the previous, default settings for retaining performance data for 13 months or to change it to 6 months. On confirming, the historical performance data after 6 months is purged.
- Before upgrading, you should close any open files or folder in *<InstallDir>\JDK* and *MySQL Data Directory*.
- If you have an active anti-virus software installed on your Windows system, Unified Manager upgrade might fail. You should disable all virus scan software on your system before upgrading Unified Manager.

During the upgrade process, Unified Manager is unavailable. You should complete any running operations before upgrading Unified Manager.

If Unified Manager is paired with an instance of OnCommand Workflow Automation, and there are new versions of software available for both products, you must disconnect the two products and then set up a new Workflow Automation connection after performing the upgrades. If you are performing an upgrade to only one of the products, then you should log into Workflow Automation after the upgrade and verify that it is still acquiring data from Unified Manager.

Steps

1. Log in to the NetApp Support Site, and navigate to the Download page for Unified Manager:

   NetApp Support Site

2. Select the required version of Unified Manager and accept the end-users license agreement (EULA).
3. Download the Unified Manager Windows installation file to a target directory on the Windows system.
4. Right-click and run the Unified Manager installer executable (.exe) file as an administrator.

   Unified Manager prompts you with the following message:

   ```
   This setup will perform an upgrade of Unified Manager. Do you want to
   continue?
   ```

5. Click **Yes**, and then click **Next**.
6. Enter the MySQL8 root password that was set during installation, and then click **Next**.
7. Log in to the Unified Manager web UI, and verify the version number.
8. If you have an active anti-virus software installed on your Windows system, make sure to manually exclude the following paths from anti-virus scan after your upgrade is complete:
   - Unified Manager data directory
   - Unified Manager installation directory
   - MySQL data directory

> To perform a silent upgrade of Unified Manager, run the following command:
> `ActiveIQUnifiedManager-<version\>.exe /s /v"MYSQL_PASSWORD=<password> /qn /l*v <system_drive>:\install.log"`

Upgrading third-party products

You can upgrade third-party products, such as JRE, on Unified Manager when installed

on Windows systems.

The companies that develop these third-party products report security vulnerabilities on a regular basis. You can upgrade to newer versions of this software at your own schedule.

Upgrading OpenJDK

You can upgrade to a newer version of OpenJDK on the Windows server on which Unified Manager is installed to obtain fixes for security vulnerabilities.

What you'll need

You must have Windows admin privileges for the system on which Unified Manager is installed.

You can update OpenJDK releases within release families. For example, you can upgrade from OpenJDK 11.0.12 to OpenJDK 11.0.13, but you cannot update directly from OpenJDK 11 to OpenJDK 12.

Steps

1. Log in as the admin user on the Unified Manager host machine.
2. Download the appropriate version of OpenJDK (64-bit) from the OpenJDK site to the target system.

   For example, download `openjdk-11_windows-x64_bin.zip from` http://jdk.java.net/11/.

3. Use the Windows Services console to stop the following Unified Manager services:
   - NetApp Active IQ Acquisition Service (Ocie-au)
   - NetApp Active IQ Management Server Service (Oncommandsvc)
4. Expand the `zip` file.
5. Copy the directories and files from the resulting `jdk` directory (for example, `jdk-11.0.13` to the location where Java is installed. Example: `C:\Program Files\NetApp\JDK\`
6. Start the Unified Manager services by using the Windows Services console:
   - NetApp Active IQ Management Server Service (Oncommandsvc)
   - NetApp Active IQ Acquisition Service (Ocie-au)

Restarting Unified Manager

You might have to restart Unified Manager after making configuration changes.

What you'll need

You must have Windows administrator privileges.

Steps

1. Log in to Windows using the default local administrator account.
2. Stop the Unified Manager services:

From the…	Stop the services in following order…
Command line	a. `sc stop ocie-au` b. `sc stop Oncommandsvc`
Microsoft Service Manager	a. NetApp Active IQ Acquisition Service (Ocie-au) b. NetApp Active IQ Management Server Service (Oncommandsvc)

3. Start the Unified Manager services:

From the…	Start the services in following order…
Command line	a. `sc start Oncommandsvc` b. `sc start ocie-au`
Microsoft Service Manager	a. NetApp Active IQ Management Server Service (Oncommandsvc) b. NetApp Active IQ Acquisition Service (Ocie-au)

Uninstalling Unified Manager

You can uninstall Unified Manager by using the Programs and Features wizard, or by performing an unattended uninstallation from the command-line interface.

What you'll need

- You must have Windows administrator privileges.
- All clusters (data sources) must be removed from the Unified Manager server before uninstalling the software.

Steps

1. Uninstall Unified Manager by choosing one of the following options:
   - If you uninstall Unified Manager from **Programs and Features** wizard, perform these steps:
     a. Navigate to **Control Panel** > **Program and Features**.
     b. Select Active IQ Unified Manager, and click **Uninstall**.
   - If you uninstall Unified Manager from Command line, then perform these steps:
     a. Log in to the Windows command line using administrator privileges.
     b. Navigate to the Active IQ Unified Manager directory, and run the following command:

     ```
     msiexec /x {A78760DB-7EC0-4305-97DB-E4A89CDFF4E1} /qn /l*v
     %systemdrive%\UmUnInstall.log
     ```

     If User Account Control (UAC) is enabled on the server, and you are logged in as a domain user,

you must use the command-line uninstallation method.

Unified Manager is uninstalled from your system.

2. Uninstall the following third-party packages and data that are not removed during the Unified Manager uninstallation:
   - Third-party packages: JRE, MySQL, Microsoft Visual C++ 2015 Redistributable, Python, and 7zip
   - MySQL application data generated by Unified Manager
   - Application logs and contents of application data directory

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